

- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

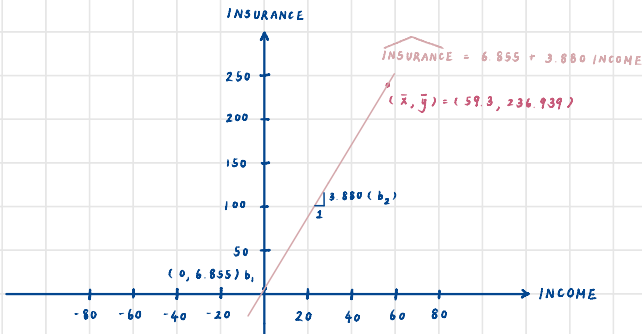
$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

3.18

a.



sample mean of income = 59.3

sample mean of the amount of insurance held = 236.939

b.

- Point estimate

the average amount of insurance held changes with each additional \$1,000 of household income
 $\Rightarrow$ the definition of slope $\Rightarrow 3.880 \times 1,000 = 3880$

- Interval estimate

degree of freedom = $N - 2 = 20 - 2 = 18$

$t_{(18, 0.975)} = 2.1009$

$se(b_2) = 0.112$

$b_2 \pm t_{(18, 0.975)} se(b_2) \Rightarrow 3880 \pm 2.1009 \times 0.112 \Rightarrow [3.645, 4.115]$

- Explain

In repeated sampling, about 95% intervals constructed this way will contain the true value of the parameter β_2 .

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

c.

$$INCOME = 100$$

$$t_{(18, 0.995)} = 2.8784$$

$$se(b_1 + 100b_2) = \sqrt{var(b_1) + var(b_2) + cov(b_1, b_2)} = \sqrt{7.383^2 + 100^2 \times 0.112 + 2 \times 100 \times (-0.746)} = \sqrt{30.749} = 5.545$$

$$b_1 + 100b_2 \pm t_{(18, 0.995)} \times se(b_1 + 100b_2) \Rightarrow 6.855 + 100 \times 3.880 \pm 2.8784 \times 5.545 \Rightarrow [378.894, 410.816]$$

d.

$$H_0: \beta_2 = 5$$

$$H_1: \beta_2 \neq 5$$

$$t^* = \frac{b_2 - c}{se(b_2)} = \frac{3.880 - 5}{0.112} = -10$$

$$\text{rejection region: } |t| > 2.1009$$

Since t^* falls in the rejection region, we reject the null hypothesis that $\beta_2 = 5$, and accept the alternative hypothesis that $\beta_2 \neq 5$.

e.

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 > 1$$

$$t^* = \frac{b_2 - c}{se(b_2)} = \frac{3.880 - 1}{0.112} = 25.714$$

$$\text{rejection region: } t > 2.5524$$

Since t^* falls in the rejection region, we reject the null hypothesis that $\beta_2 = 1$, and accept the alternative hypothesis that $\beta_2 > 1$. At the 1% significant level, we have statistical evidence that as income increases, the increase in the amount of life insurance held is greater than the increase in income.

3.23

a.

- $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$ marginal effect = $2\alpha_2 \overset{20}{SQFT}$
- t-statistic: $t^* = -26.7287$ $\frac{2 \times 20 \times d_2 - 13}{2 \times 20 \times 5e}$
- rejection region: $\{t \mid t > t_{(0.95, 498)} \approx 1.648\}$
- p-value: $P(T > t^*) = 1$
- Conclusion:

★ Hypothesis

$$H_0: 2 \times 20 \times d_2 = 13$$

$$H_1: 2 \times 20 \times d_2 > 13$$

- Since the p-value is greater than 0.05, we do not reject the null hypothesis. This implies that adding 100 square feet to a 2,000-square-foot house does not significantly increase its expected price by more than \$13,000.

b.

- $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$ marginal effect = $2\alpha_2 \overset{40}{SQFT}$
- t-statistic: $t^* = 4.1895$ $\frac{2 \times 40 \times d_2 - 13}{2 \times 40 \times 5e}$
- rejection region: $\{t \mid t > t_{(0.95, 498)} \approx 1.648\}$
- p-value: $P(T > t^*) = 165e - 5$
- Conclusion:

★ Hypothesis

$$H_0: 2 \times 40 \times d_2 = 13$$

$$H_1: 2 \times 40 \times d_2 > 13$$

- Since the p-value is less than 0.05, we reject the null hypothesis. This implies that adding 100 square feet to a 4,000-square-foot house significantly increases its expected price by more than \$13,000.

c.

- $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$
- $E(PRICE \mid SQFT = 20) = 167.3735$
- 95% Confidence Interval of the expected price = [158.0481, 176.6988]
- Conclusion:

- In repeated sampling, about 95% intervals constructed this way will contain the true value of the expected price.

d.

- The sample mean is 163.3333.
- The sample average of 163.3333 falls within the interval, which is compatible with the results in part c.

3.31

a.

- SAL1

- Mean = 6719
- Minimum = 1762
- Maximum = 32820
- Standard deviation = 6915.083

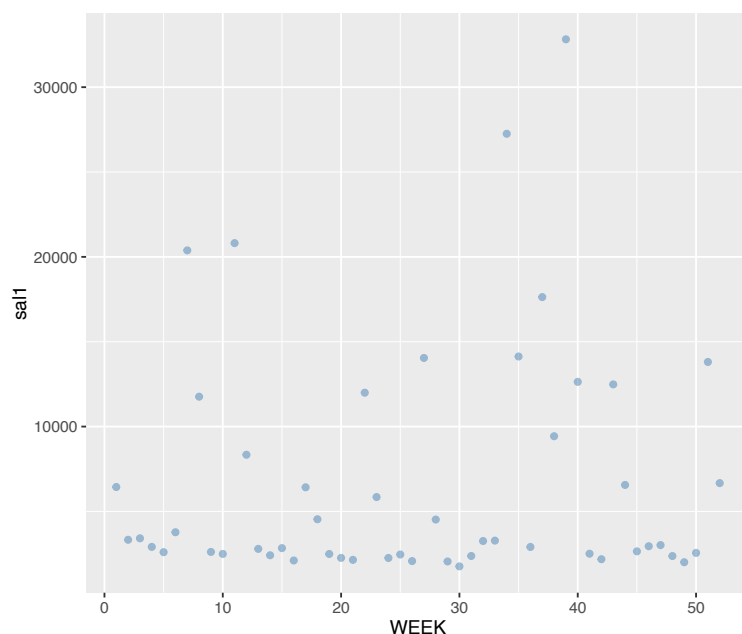
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> # part a
> summary(tuna$sal1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1762   2485   3136   6719   8612  32820

> summary(tuna$apr1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
```

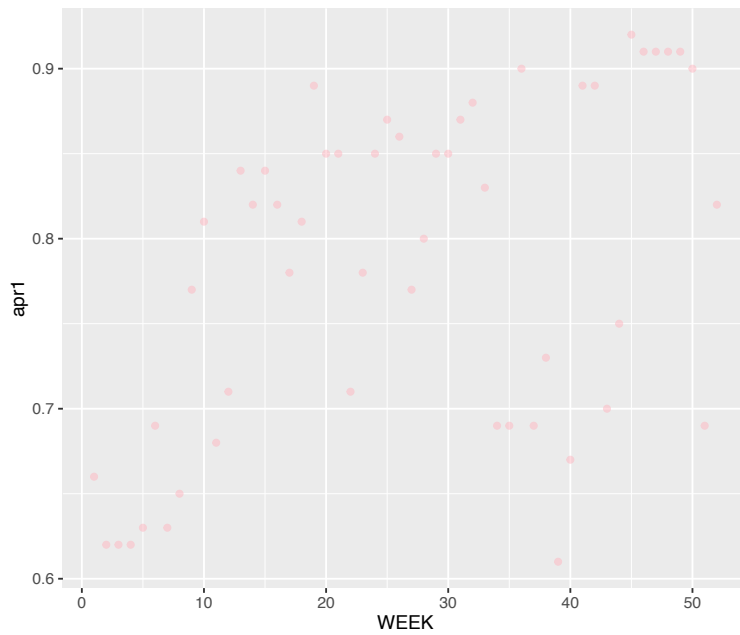
- APR1

- Mean = 0.7825
- Minimum = 0.6100
- Maximum = 0.9200
- Standard deviation = 0.098

```
> sd(tuna$sal1)
[1] 6915.083
> sd(tuna$apr1)
[1] 0.09803711
```

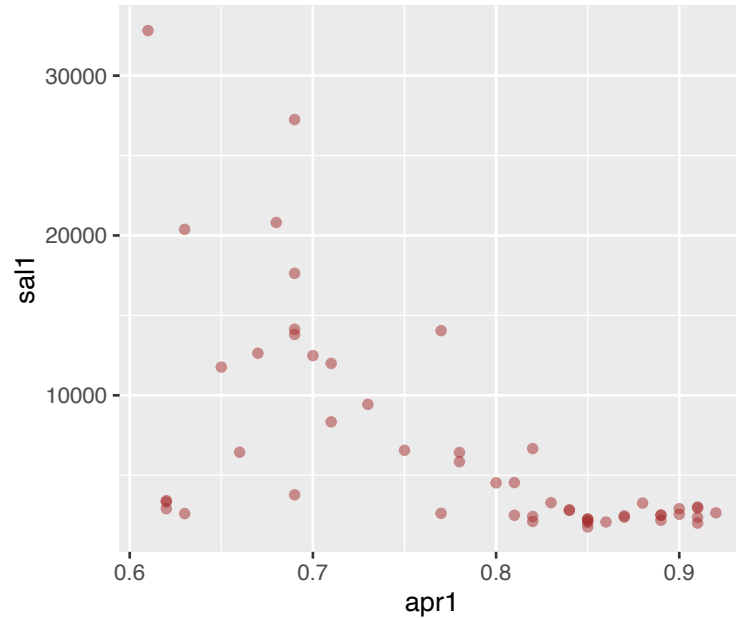


- The variation in sales is **highly volatile**. For most of the weeks, sales hover at a relatively low baseline. However, this baseline is punctuated by several dramatic, isolated spikes where sales suddenly surge to roughly 20,000. This indicates that sales are far from stable.



- The price **fluctuates significantly and continuously** from week to week. Rather than remaining stable, the price constantly bounces up and down. This shows a highly dramatic weekly pricing pattern.

b.



- There is **inverse relationship**. As the price increases, the sales volume tends to decrease. This aligns with the law of demand.

c.

- $SAL1 = \beta_1 + \beta_2 \text{ PRICE1} + e$ (in cents) PRICE 1 = 100 APR 1
- Point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1 = **-434** (β_2)
- 95% Confidence Interval for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1 = [-592.2897, -276.6049]

d.

- The price per can is 70 cents.
- 90% Confidence Interval for the expected number of cans sold in a week = [8624.934, 11980.87]

e.

- $e = -5.06$ elasticity = $\beta_2 \times \frac{\text{price 1}}{\text{sal 1}}$
- 95% Confidence Interval of elasticity of sales with respect to the price = [-6.898, -3.222]
- The sales are **elastic** because the absolute value of the elasticity is greater than 1. This means that a 1% increase in price leads to a massive 5.06% drop in sales. It **makes economic sense**. Canned tuna is a common product with many **similar substitutes**. Because it is easy to switch between different brands, consumers are highly sensitive to price.

f.

- $H_0: e = -3$ vs. $H_1: e \neq -3$
- t-statistic: $t^* = -2.25$ $se(e) = se(\beta_2) \times \frac{\text{price 1}}{\text{sal 1}}$
- rejection region: $\{t \mid t > t_{(0.95, 48)} \approx 1.6759\}$
- p-value: $2 \times P(T > |t^*|) = 0.0288$
- Conclusion:
 - Since the **p-value is less than 0.1**, we **reject the null hypothesis**. This implies that the elasticity of sales with respect to price is significantly different from -3.